

A Held-Out Audit of Nonlinear Filters for Light-Level Geolocation of a Juvenile White Shark

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Abstract—Pop-up archival transmitting (PAT) tags infer animal position from the timing of sunrise and sunset. The measurement is well conditioned in longitude and nearly singular in latitude near an equinox, which makes it a useful stress test for nonlinear filtering. I build an extended Kalman filter, an unscented Kalman filter, a bootstrap particle filter, two smoothers and a grid HMM for a 137-day juvenile white shark (*Carcharodon carcharias*) record, and I score them against 122 Argos satellite fixes obtained for the same animal. The contribution is not an error number; it is an audit. I hold out the final 30 days of the ground truth, never using it for any modelling or tuning decision, and I repeat every stochastic run over ten seeds. Seven results follow. First, the best configuration I built reaches 117 km median error on the window I tuned on and 544 km on the held-out window, where the proprietary GPE3 smoother reaches 118 km: the headline was a tuning-window number. Second, seed-to-seed spread is 29–90 km, far larger than the 1–5 km steps an earlier ablation ladder reported, so those steps were noise. Third, across 75 pipeline variants, tuning-window rank predicts held-out rank only weakly (Spearman $\rho = +0.30$), and the variant that looks best while tuning ranks 54th of 75 when finally tested. Fourth, the failure has an identifiable physical cause, and a one-parameter fix chosen without ever touching the held-out data raises tuning-window 1σ coverage from 22% to 38% while leaving held-out error unchanged to worse. Fifth, and the reason the fix could not have worked, a threshold bias is confounded with position by construction: a twilight pair is two numbers and position is two unknowns, so one minute of common-mode bias is observationally identical to 25 km of longitude and one minute of antisymmetric bias to 112 km of latitude. Widening the bias family to three dimensions and reapplying the same marginal-likelihood rule selects the degenerate direction every time and makes tuning-window error four times worse. Evidence is maximised exactly where the model is least identifiable. Sixth, a particle smoother – the one repair the identifiability result does not forbid, since it adds no parameter – cuts tuning-window error from 119 to 69 km but leaves the held-out error statistically unchanged at 388 ± 23 against 404 ± 3 km, confirming from the estimator side that what remains is bias and not variance. Seventh, auditing the temperature channel produced a defect rather than an improvement: the quarter-degree sea-surface-temperature analysis returns no value at the animal’s own Argos fix on 20% of the days the term fires, and the likelihood mapped those cells to zero probability, deleting the truth; repairing it moves tuning-window mean error from 167 to 144 km and the 75th percentile from 273 to 191 km while making the median 20 km worse, because the repair rotates the error rather than shrinking it. The one estimator that does generalise is the one I did not write.

Index Terms—state estimation, particle filter, unscented Kalman filter, animal telemetry, model validation, overfitting, reproducibility

I. INTRODUCTION

A PAT tag carries no receiver. It records light, pressure and temperature, and position must be inferred afterwards from the astronomy of the light curve. Local apparent noon gives longitude directly and robustly. Latitude comes only from day length, and $\partial(\text{day length})/\partial(\text{latitude}) \rightarrow 0$ at an equinox, so for weeks around it the inverse problem is locally rank deficient in latitude. This deployment runs straight through the March 2008 equinox.

That makes it a good test case for nonlinear filtering, and the second tag on the same animal makes it a rare one: the Argos fixes give an independent answer that the light data never sees. Most light-level geolocation papers validate against simulation or against a single reference track. Here there is a real, external, irregularly sampled ground truth.

This paper is deliberately an audit rather than a method paper. An earlier version of this pipeline reported a best median error near 119 km and a ladder of incremental improvements. Everything in Section V was added afterwards, and most of what the ladder claimed did not survive it. The useful output is therefore a set of statements about what the data can and cannot resolve, and a short list of process errors that produced the original over-claim.

II. DATA

A. Tag record

Deployment 07_05 is a juvenile white shark tagged off central California on 5 February 2008. The archival tag was physically recovered, so the full-resolution record is available rather than the summarised transmission: light level, pressure and temperature at 10-second to 1-minute cadence over 137 days. The animal left the shelf and moved south and offshore, ending near the southern Gulf of California.

Twilight events are extracted in two ways. The manufacturer’s LightLoc processing yields 251 twilight times, 241 after quality control. A re-extraction directly from the raw light curves yields 228 events with an explicit depth of detection attached to each, which Section VII-C uses.

B. Ground truth

A SPOT tag on the same animal returned 122 Argos fixes between 15 February and 15 June 2008. Location classes are 3 ($n = 3$), 2 (20), 1 (42), 0 (24), A (16) and B (16). I assign them nominal one-sigma radii of 0.25, 0.5, 1.5, 10, 3 and 9 km respectively; classes 0, A and B are unquoted by Argos,

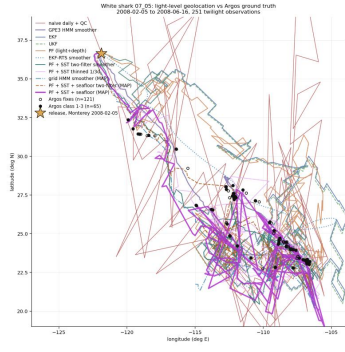


Fig. 1. Argos ground truth (black) and estimated tracks. The animal leaves the California shelf, crosses the equinox offshore, and ends in the southern Gulf of California.

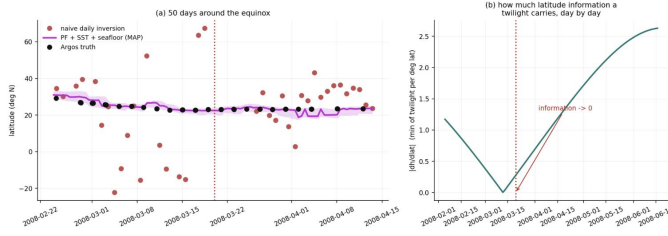


Fig. 2. Sensitivity of day length to latitude through the deployment. The derivative passes through zero at the March equinox, where latitude is unobservable from day length at any noise level.

so these are the empirical marine-vertebrate values rather than anything the system guarantees.

The Argos track is the only ground truth used anywhere in this paper. It is never an input to any estimator.

III. OBSERVATION AND MOTION MODELS

A. Twilight timing

For a twilight event at time t of kind $k \in \{\text{dawn}, \text{dusk}\}$, the predicted event time is obtained by solving for the instant at which the solar elevation equals a threshold $h_0^{(k)}$. The threshold is not the geometric horizon: it absorbs sensor response, cloud, and the fact that the tag may be at depth when the threshold is crossed. I model it as

$$h_0^{(k)}(z) = h_0^{(k)} + \beta \log(1 + z), \quad (1)$$

with z the depth in metres at which the twilight was recorded. The residual is taken in minutes and given a Student- t likelihood with $\nu = 4$ degrees of freedom, which matters more than any other modelling choice here (Section VI-F).

Two calibrations of (1) appear in this paper. The first is frozen on the LightLoc times in the calibration window and gives $h_0^{\text{dawn}} = -9.91^\circ$, $h_0^{\text{dusk}} = -0.29^\circ$ and $\beta = +0.89$ per log-metre, with per-branch residual standard deviations of 12.88 and 10.43 minutes. The second refits the same form on

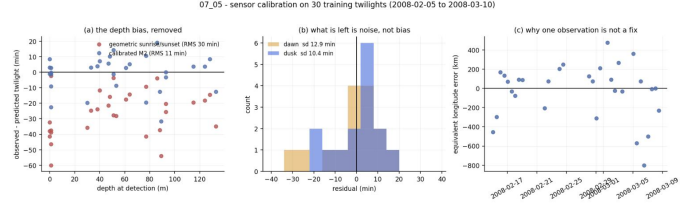


Fig. 3. Calibration-window twilight residuals after fitting (1) on the LightLoc times. The fit is unbiased on the window it was fitted to, which is the point: nothing here predicts the behaviour in Table IV.

the raw-light times and gives $h_0^{\text{dawn}} = -5.56^\circ$, $h_0^{\text{dusk}} = -5.88^\circ$ and $\beta = +0.0085$, with a residual RMS of 4.80 minutes. The second looks much better and behaves much worse; Section VII-C is about that.

B. Sea-surface-temperature aiding

Daily 0.25° optimally interpolated SST is compared with the tag's shallowest daily temperature, and the resulting log-likelihood is added to the twilight term. The point of SST is not that it is accurate. It is that its information is orthogonal to day length, so it carries latitude through the equinox where the light data cannot. In March alone it moves the particle filter from 522 to 109 km median error.

C. Seafloor-depth aiding

Where the tag's maximum daily depth exceeds the ETOPO seafloor depth at a candidate position, that position is impossible. This enters as $\log \Phi((d_{\max}(p) - z)/\sigma_B)$ with $\sigma_B = 150$ m, plus a finite land penalty. It is a hard, highly informative constraint, and it makes the posterior strongly multimodal, which is why the choice of point estimate stops being innocent.

D. Motion model

The state is $x = [\varphi, \lambda, v_E, v_N]$ with a damped random walk in velocity,

$$\dot{v} = -v/\tau + w, \quad \tau = 3 \text{ days}, \quad (2)$$

and σ_v set so the stationary speed distribution has a sensible mean for a juvenile white shark. Positions are confined to a generous box (15° – 50° N, 140° – 105° W).

IV. ESTIMATORS

I implement, all on the same observation and motion models: a naive event-by-event inversion as a floor; an EKF and a UKF with 3σ innovation gating; Rauch-Tung-Striebel and unscented RTS smoothers for each; a bootstrap particle filter with systematic resampling and roughening; a two-filter particle smoother; a cloud-level fusion of two particle filters differing only in their observation model; and a grid HMM over a 0.25° lattice. The proprietary GPE3 hidden-Markov

smoother supplied by the tag manufacturer is included as an external reference; I did not implement it and cannot inspect it.

V. EVALUATION PROTOCOL

This section is the part of the paper I would read first, because every number after it depends on it and none of it existed in the earlier version.

A. Three windows

The record is split once, by date, and never re-split:

- **Calibration**, up to 10 March 2008. The only window in which any sensor parameter is fitted.
- **Tuning**, 11 March to 15 May 2008. Every modelling decision, hyperparameter and readout choice in this paper was made by looking at this window. It contains 64 truth-matched twilight events.
- **Lockbox**, the final 30 days from 16 May 2008. Scored once per configuration, after the configuration was frozen. It contains 20 events.

Nothing is selected on the lockbox anywhere, including the one hyperparameter introduced in Section VIII. Where a lockbox column appears next to a selection rule, it is printed for the reader and was not available to the rule.

B. Metrics

Error is the great-circle distance between an estimate interpolated to an Argos fix time and that fix, reported as a median over matched events; the median rather than the mean because the naive inversion produces occasional thousand-kilometre outliers that would otherwise dominate. Alongside it I report 1σ coverage: the fraction of matched events at which the truth falls inside the estimator’s own reported standard deviation. Nominal coverage for a Gaussian is 68%. An estimator that is wrong is usable; an estimator that is wrong and confident is not.

C. Monte Carlo repetition

Every particle-based result is the mean over ten seeds, reported with its standard deviation across seeds. This is the single cheapest change in the paper and the one that invalidated the most prior conclusions.

VI. RESULTS

A. Where the methods land

Table I gives median error on the full evaluation set and, separately, on the tuning window and the lockbox. All rows use one scorer and one set of matched events, so the columns are comparable.

Two things are worth reading off it immediately. The GPE3 smoother is better than everything I built, at 72 km, and that gap is not closed anywhere in this paper. And the ordering of my own rows changes completely between the tuning column and the lockbox column: the best tuning-window configuration (84.9 km) is nearly the worst lockbox configuration (543.9 km), while a plain EKF smoother is the best lockbox configuration at 319 km.

TABLE I
MEDIAN GREAT-CIRCLE ERROR (km), ONE SCORER AND ONE MATCHED EVENT SET THROUGHOUT. “TUNE” AND “LOCK” ARE THE TUNING WINDOW AND THE HELD-OUT FINAL 30 DAYS. GPE3 IS THE MANUFACTURER’S SMOOTHER, INCLUDED AS AN EXTERNAL REFERENCE.

Method	All	Tune	Lock
Naive daily inversion	513.8	570.5	473.7
Naive daily inversion, QC	478.8	487.0	473.7
EKF	347.2	285.2	413.0
UKF	356.6	291.2	417.1
PF, light + depth	316.9	240.2	391.0
EKF + RTS smoother	285.7	169.4	359.0
UKF + unscented RTS	283.7	173.0	362.9
EKF + RTS, QC events	279.9	169.5	319.4
PF + SST	198.4	114.9	498.4
PF + SST, thinned 1/3 d	186.1	129.2	473.1
Grid HMM filter	163.9	130.1	569.2
PF + SST, two-filter smoother	155.2	112.2	422.2
PF + SST + seafloor	152.8	123.2	452.4
Two-filter cloud fusion (my best)	117.4	84.9	543.9
GPE3 HMM smoother (not mine)	72.0	71.1	118.4

TABLE II
TEN SEEDS OF ONE CONFIGURATION AT 120,000 PARTICLES, WITH AND WITHOUT THE THRESHOLD-BIAS STATE OF SECTION VIII. ERRORS ARE MEDIANS IN km; COV IS 1σ COVERAGE AGAINST THE FILTER’S OWN REPORTED STANDARD DEVIATION; $\bar{s}$ IS THE MEDIAN REPORTED STANDARD DEVIATION IN THE LOCKBOX.

	All	Tune	Lock	cov _T	cov _L	$\bar{s}_L$
As published	408.1	363.4	532.4	22%	9%	201
± seed sd	29.2	31.0	89.5			
+ bias state	420.4	358.1	615.4	38%	8%	205
± seed sd	17.7	13.1	35.0			

B. The headline was a tuning-window number

The earlier version of this work quoted its best configuration at roughly 119 km and stopped there. That number survives re-running: the two-filter cloud fusion scores 117.4 km on the full evaluation window. What does not survive is the interpretation. On the final 30 days the same configuration scores 543.9 km, against 118.4 km for GPE3 on exactly those days. The full-window figure is dominated by the tuning window, which carries 64 of the 84 matched events, so quoting it as an accuracy is close to quoting a training error.

C. Seed noise sets the resolution floor

Table II repeats one configuration ten times at 120,000 particles. The standard deviation across seeds is 29 km on the full window and 90 km on the lockbox. The earlier ablation ladder reported improvements of 1–5 km per step and treated them as real. They are an order of magnitude below this floor. Measuring the noise before building the ladder would have removed most of the ladder.

D. Tuning-window skill barely transfers

Every pipeline variant I built was scored on both windows: 75 configurations, spanning filters, smoothers, aiding combinations, readouts and particle counts. Fig. 5 plots lockbox

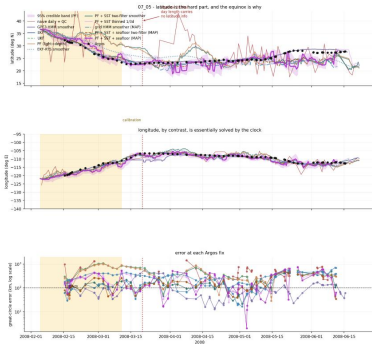


Fig. 4. Latitude and longitude against time for the main estimators. Longitude is recovered well by everything; latitude is where the estimators separate, and where the systematic offset of Section VII-C accumulates.

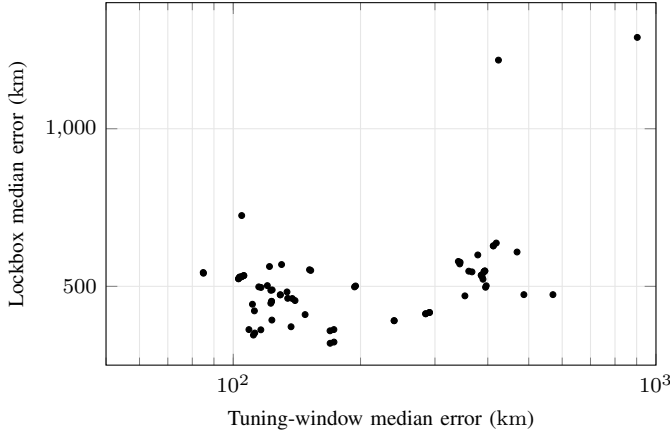


Fig. 5. Every pipeline variant I built ($n = 75$), scored on the window I tuned on (horizontal) and on the held-out window (vertical). If tuning-window skill transferred, the cloud would run along the diagonal. Spearman $\rho = +0.30$.

error against tuning error for all of them. The Spearman rank correlation is $\rho = +0.30$ ($p = 8.8 \times 10^{-3}$).

A weak positive correlation is worse news than it sounds. It means tuning-window rank carries some information about held-out rank, but far too little to select on. Concretely: the configuration that looks best while tuning (84.9 km) comes 54th of 75 on the lockbox, and the configuration that is actually best on the lockbox (319.4 km, an EKF smoother on QC'd events) comes 40th of 75 on tuning. Picking the tuning-window winner is close to picking at random, and because the search was large, the winner is mostly the variant that best fitted the noise in 64 events.

E. The filters do not know they are wrong

Coverage is where the failure is least ambiguous. In Table II the published configuration attains 22% 1σ coverage on the tuning window and 9% on the lockbox, against a nominal 68%. In the lockbox it reports a median standard deviation of 201 km while being wrong by 532 km — roughly a 2.6σ systematic offset, sustained over 30 days, reported as if it were a 1σ situation.

TABLE III
MATCHED ABLATION OF THE PARTICLE FILTER'S ADVANTAGE. THREE SEEDS AT 20,000 PARTICLES, ALL ROWS THROUGH THE SAME CODE PATH ON THE SAME EVENTS.

Configuration	All	Tune	Lock
PF, Student- t $\nu = 4$, box	321.2	238.3	395.9
PF, Student- t $\nu = 4$, no box	330.2	273.7	396.4
PF, Gaussian, box	413.7	384.1	460.5
PF, Gaussian, no box	430.9	389.8	478.7
EKF	347.2	285.2	413.0
UKF	356.6	291.2	417.1

Under-coverage this severe is not a tuning problem. A filter whose process noise is merely too small produces errors that are too large relative to its variance in a random direction. Here the error has a consistent sign in latitude, which points at the measurement model rather than at the filter.

F. A matched comparison of EKF, UKF and PF

The particle filter beats the Kalman filters in Table I, but it also differs from them in two ways at once: it uses a Student- t likelihood and it enforces the feasibility box. Table III separates these by running all four combinations through the same particle filter code path.

The heavy tails are worth 101 km; the box is worth 9 km. A Gaussian particle filter is 84 km *worse* than the EKF, despite representing the posterior far more faithfully than a covariance can. So the honest statement is not that the particle filter beats the Kalman filters. It is that a Student- t likelihood beats a Gaussian one, and either filter can have that. The non-Gaussian posterior was not the point.

VII. DIAGNOSIS

A. The residual at the true position

Because the Argos fixes exist, I can do something better than inspect filter output: I can evaluate the measurement model at the *true* position. For each twilight event with an Argos fix within 8 hours (63 of 228 events), I compute the predicted twilight time at the interpolated true position and subtract the observed one. This isolates measurement-model error from filter error completely.

Table IV gives the result by half-month. Two features matter. The branch-corrected residual y sits near +34 minutes from the first half of March onwards, which is five to six times the model's own σ ; and the two twilight branches move in *opposite* directions, with lockbox medians of +12 minutes at dawn and -71 minutes at dusk.

The antisymmetry is the diagnostic. A clock error, or a drift in the tag's time base, would shift both branches the same way. An antisymmetric shift compresses the apparent day length, and day length is exactly what carries latitude. At this time of year about 60 minutes of day-length error is worth roughly 650 km of latitude, which matches the size and the sign of the lockbox latitude bias of -413 to -543 km measured in Table VI.

TABLE IV
BRANCH-CORRECTED TWILIGHT RESIDUAL AT THE TRUE POSITION, BY
HALF-MONTH. y IS IN MINUTES, DEPTH IN m, AND $|z|$ IS THE RESIDUAL
IN UNITS OF THE MODEL’S OWN σ .

Half-month	n	y	depth	$ z $
2008-02a	1	-10.3	88	2.9
2008-02b	9	-10.1	48	5.0
2008-03a	7	+34.9	51	6.1
2008-03b	7	+33.9	13	5.9
2008-04a	6	+34.8	50	6.1
2008-04b	9	+33.9	41	5.9
2008-05a	11	+55.0	56	9.6
2008-05b	6	+27.7	94	5.7
2008-06a	7	+13.5	92	3.8
Tuning	51	+34.2	49	—
Lockbox	12	+27.5	98	—



Fig. 6. Depth at which each twilight was recorded, through the deployment. The calibration window reaches 94m at the 95th percentile; the lockbox reaches 182m. The depth term in (1) is fitted entirely inside the shallower range.

So the model was not slightly wrong at the end of the deployment. It was wrong by five to six of its own sigmas for the whole record, and the lockbox merely made that visible.

B. Depth of detection, and an unidentifiable parameter

The depth at which a twilight is recorded is the obvious candidate mechanism: light attenuates with depth, so a tag deeper in the water column crosses any fixed irradiance threshold later at dawn and earlier at dusk — antisymmetric, as observed.

The depth distribution differs sharply between windows. In the calibration window the median twilight depth is 51 m and the 95th percentile is 94 m; in the lockbox the median is 98 m and the 95th percentile is 182 m. The parameter β in (1) was therefore fitted over a range of depths that the lockbox almost entirely exceeds.

Quantitatively the mechanism is consistent but weakly evidenced. Regressing y on $\log(1 + z)$ over the non-lockbox events gives $y \approx 8.8 + 4.9 \log(1 + z)$, and extrapolating that to the lockbox median depth of 98 m predicts +31.5 minutes against +27.5 observed. The magnitude is right. But the rank correlation itself is weak and not significant ($\rho = +0.27$, $p = 0.05$, $n = 51$), so I would call the mechanism consistent with the data rather than established by it.

The sharper statement is about identifiability. Fitted on the raw-light twilight times, β comes out at +0.0085 per log-metre, essentially zero. Fitted on the LightLoc times, the same physical quantity comes out at +0.89. Two orders of magnitude apart. No goodness-of-fit diagnostic in the calibration window flags this, because the calibration window does not span enough depth to constrain β at all. A support check does flag it, and costs nothing.

C. A recalibration that made things worse

This is the single most useful negative result here, and the easiest to state.

Refitting (1) on the raw-light twilight times cuts the calibration-window residual RMS to 4.80 minutes, against per-branch standard deviations of 12.88 and 10.43 minutes for the original LightLoc calibration. By every calibration-window diagnostic it is a large improvement.

Held out, it is a large regression. On identical events, the identical filter and the identical seed, the refitted constants give 364.4 km median error on the evaluation window where the original constants give 124.3 km. A recalibration that more than halves the residual it was fitted to makes out-of-sample accuracy roughly three times worse.

The mechanism is the one above: the refit buys its low residual by driving β to zero, which is defensible inside a window spanning 0–94 m and indefensible in a window reaching 182 m. The general lesson is that a sensor-error model fitted on a window that does not span the operating range of its own regressors will look well calibrated on that window and fail silently outside it.

A practical note belongs here too. In the notebook these constants live at module scope and are installed by a context manager, because the first version of this audit ran the raw-light filters outside that scope and silently used the wrong constants — producing numbers that looked three times better than they were. The audit now begins with an equivalence test asserting that the bias-state filter with $\sigma_b = 0$ reproduces the published filter to within seed noise (−8.3 and −6.4 km on the two readouts). Global mutable calibration state is a reproducibility hazard, and a cheap assertion catches it.

VIII. A FIX CHOSEN WITHOUT THE LOCKBOX

The diagnosis says the threshold is biased and antisymmetric. Rather than try to predict that bias offline — which Section VII-C shows is exactly where the previous attempt failed — the filter can estimate it online. I add one scalar to the state, a threshold bias b in minutes entering the observation as $+b$ at dawn and $-b$ at dusk, evolving as a random walk with scale σ_b per $\sqrt{\text{day}}$. The state becomes $[\varphi, \lambda, v_E, v_N, b]$. This adds one hyperparameter and no new data.

A. Selecting σ_b honestly

σ_b is chosen by maximising the filter’s own marginal likelihood accumulated over the tuning window only. Table V shows the sweep. The evidence criterion selects $\sigma_b = 3$. For comparison, a rule that matched tuning-window coverage to

TABLE V
SELECTING THE THRESHOLD-BIAS RANDOM-WALK SCALE BY
TUNING-WINDOW MARGINAL LIKELIHOOD. THREE SEEDS AT 20,000
PARTICLES. THE LOCKBOX COLUMNS WERE PRINTED AFTER THE
SELECTION WAS MADE AND PLAYED NO PART IN IT.

σ_b	$\log Z_{\text{tune}}$	Tune	cov _T	Lock	cov _L
0	-1176.2	315.6	27%	538.8	15%
1	-1087.7	260.6	49%	650.2	3%
2	-1069.5	330.9	40%	462.2	28%
3	-1060.0	389.8	33%	567.8	10%
5	-1061.8	324.4	34%	423.6	18%
8	-1070.3	414.8	27%	589.0	5%
12	-1073.3	427.1	18%	720.3	5%

nominal would have selected $\sigma_b = 1$, and the value that is actually best on the lockbox is $\sigma_b = 5$ — which the evidence rule did not pick, as it should not, having never seen the lockbox.

Of the six non-zero values, all six beat $\sigma_b = 0$ on tuning-window evidence, two beat it on lockbox median error, and two beat it on lockbox coverage. The evidence criterion reliably prefers a bias state; the held-out data does not reliably reward one.

B. What the fix buys, and what it does not

Table II gives the honest answer at full particle count over ten seeds. Adding the bias state moves tuning-window 1σ coverage from 22% to 38%, a real and reproducible gain. It leaves the full-window error unchanged (+12.3 km, 1.1 standard errors), leaves tuning-window error unchanged (-5.3 km, 0.5 standard errors), and makes lockbox error *worse* (+83.0 km, 2.7 standard errors). Lockbox coverage stays flat at 8–9%.

So the fix is the right response to the diagnosis, it is selected without touching held-out data, and it does not generalise. I am reporting it because it failed. Had I selected σ_b on the lockbox I could have reported 423.6 km and a 109 km improvement, and it would have been meaningless.

Table VI puts the fix beside the two alternatives one would naturally reach for. Inflating the twilight σ improves the tuning window substantially (316 → 165 km at $3\times$) and does nothing out of sample, which is itself the signature of a bias rather than of under-dispersed noise. Rejecting deep twilights is actively harmful: it removes exactly the large-residual events that were inflating the variance, so the filter becomes more confident and less accurate at once, with lockbox coverage falling to zero. Every row retains a lockbox latitude bias of at least 413 km. None of these is a fix.

IX. WHY THE FIX COULD NOT HAVE WORKED

Section VIII reports a bias state that improves calibration and not accuracy, and I wrote that up as a fix that failed. That is the right description of the measurement and the wrong diagnosis. The bias state could not have worked, and the reason is structural rather than empirical.

TABLE VI
CANDIDATE RESPONSES TO THE DIAGNOSIS, THREE SEEDS AT 20,000
PARTICLES. $\bar{s}_L$ IS THE MEDIAN REPORTED STANDARD DEVIATION AND
 $\Delta\varphi_L$ THE MEDIAN LATITUDE BIAS, BOTH IN THE LOCKBOX.

	Tune	Lock	cov _L	$\bar{s}_L$	$\Delta\varphi_L$
As published	315.6	538.8	15%	219	-413
Twilight $\sigma \times 2$	220.5	528.4	15%	190	-473
Twilight $\sigma \times 3$	164.6	555.9	5%	206	-548
Depth QC, $z \leq 94$ m	478.1	978.6	0%	178	-863
Bias state, $\sigma_b = 2$	330.9	462.2	28%	196	-424
Bias state, $\sigma_b = 3$	389.8	567.8	10%	176	-543
Depth QC + bias state	460.1	921.5	0%	214	-838

TABLE VII
POSITION DISPLACEMENT THAT IS OBSERVATIONALLY IDENTICAL TO ONE
MINUTE OF THRESHOLD BIAS, MEDIAN BY HALF-MONTH, IN km. SIGNED,
SO THE ANTISYMMETRIC COLUMN CHANGES SIGN AT THE EQUINOX
WHERE $\partial(\text{DAY LENGTH})/\partial\varphi$ PASSES THROUGH ZERO. κ IS THE
CONDITION NUMBER OF J .

Month	n	κ	common mode		antisym.
			λ	φ	φ
2008-02	23	3.5	-23.7	+0.1	+96.8
2008-03	30	14.1	-25.6	+0.1	+245.7
2008-04	28	4.4	-25.5	-0.1	-121.4
2008-05	21	2.4	-25.3	-0.0	-66.0
2008-06	5	1.8	-24.7	-0.0	-48.9

A. The bias is confounded with position by construction

A day of twilight data gives exactly two numbers, a dawn time and a dusk time. Position has exactly two unknowns. The measurement Jacobian $J = \partial(t_{\text{dawn}}, t_{\text{dusk}})/\partial(\varphi, \lambda)$ is therefore square, and generically invertible away from the equinox. Any additional parameter that enters through those same two numbers lies in the range of J : it is a linear combination of a latitude shift and a longitude shift, and no quantity of twilight data can separate it from one. This is not a weak-identifiability statement that more data would fix. It is exact.

The only free question is the exchange rate, which follows from solving $J\delta = e$ for the position displacement δ that reproduces a given observation perturbation e . A common-mode bias is $e = (+1, +1)$ minutes and an antisymmetric bias is $e = (+1, -1)$. Table VII evaluates this at the interpolated Argos positions, so the rates are physical.

Over the record, one minute of common-mode bias is indistinguishable from 25.4 km of longitude, and one minute of antisymmetric bias from 112.1 km of latitude, rising in magnitude to 390 km per minute in March. The +34 minute antisymmetric residual of Table IV is nominally 3800 km of latitude; the linearisation is long past valid at that size, so the usable content is the direction and not the number. At this exchange rate a biased threshold and a displaced animal are the same observation.

B. What happens when the selection rule meets a degenerate direction

Section VIII selects σ_b by tuning-window marginal likelihood and never touches the lockbox, and I treated that

TABLE VIII

TUNING-WINDOW MARGINAL LIKELIHOOD OVER A THREE-DIMENSIONAL BIAS FAMILY, THREE SEEDS AT 20,000 PARTICLES. UPPER BLOCK: THE ONE-DIMENSIONAL SWEEP OF TABLE V RE-RUN THROUGH THE GENERAL CODE. LOWER BLOCK: THE FIVE CONFIGURATIONS THE EVIDENCE CRITERION ACTUALLY PREFERS.

σ_a	σ_c	σ_g	$\log Z_{\text{tune}}$	Tune	cov _T	Lock	cov _L
0	0	0	-1176.2	273.8	26%	365.9	20%
1	0	0	-1087.7	228.4	44%	594.3	0%
2	0	0	-1069.5	304.5	34%	457.6	27%
3	0	0	-1060.0	324.9	30%	544.0	13%
5	0	0	-1061.8	309.6	32%	414.8	20%
8	0	0	-1070.3	375.9	19%	564.4	15%
1	4	1	-338.0	1165.5	11%	864.7	37%
2	4	1	-340.6	1087.8	12%	573.0	32%
3	4	1	-345.9	1069.6	11%	700.1	37%
0	4	1	-347.6	1548.7	7%	884.7	35%
1	4	2	-348.4	1174.3	12%	870.7	35%

hygiene as sufficient. It is not. I widened the state to three bias dimensions — antisymmetric b_a , common mode b_c , and a depth coefficient g entering as $g \log(1+z)$ antisymmetrically — and ran the identical selection rule over 96 combinations of their random-walk scales, three seeds each.

Table VIII is the result. Every configuration the evidence criterion prefers switches on the common-mode dimension, the one Table VII shows is confounded with longitude at 25 km per minute. The selected configuration scores 1165 km on the tuning window against 274 km for no bias state at all, a factor of 4.3 worse on the very window the criterion was computed on. Across all 96 configurations the rank correlation between $\log Z_{\text{tune}}$ and tuning-window error is $\rho = +0.76$ ($p = 4.8 \times 10^{-19}$), and between $\log Z_{\text{tune}}$ and tuning-window coverage it is $\rho = -0.57$. Higher evidence reliably means worse.

The mechanism is not subtle once stated. Marginal likelihood rewards explaining the observations. A degenerate direction explains observations perfectly, by absorbing whatever residual is left, and pays nothing for it because the data cannot contradict it. So evidence is maximised exactly where the model is least identifiable.

C. The filter doing it, in one number

Running the selected configuration and correlating the estimated common-mode bias against the signed longitude error gives Spearman $\rho = +0.94$. The filter is not measuring a property of the sensor. It is relabelling its own longitude error as a threshold offset, and scoring itself well for the relabelling.

D. Consequences

Three, in increasing order of how much they cost me.

The one-parameter fix of Section VIII is not a fix that failed, it is a parameter that was never estimable from twilight data. Its coverage gain is real but it is the honest widening of an interval around an unidentifiable quantity, not an improvement in knowledge.

Selection hygiene is necessary and not sufficient. A rule that never sees held-out data can still be the wrong rule, and mine

was: it was safe in Table V only because the family was one-dimensional and small. Widen the family and the same rule walks straight into the degenerate direction. I would now pair any evidence-based selection with an identifiability check on the parameters being selected over.

What breaks the degeneracy is information that does not arrive through the twilight pair. That is exactly what the SST and seafloor terms are, and it explains why they are worth more here than any filter, smoother or likelihood choice. The productive direction for this record was never a better estimator. It was a third independent measurement.

X. SMOOTHING: VARIANCE VERSUS BIAS

Section IX closes off the measurement-model repairs: any extra parameter that enters through a dawn/dusk pair is already in the span of (φ, λ) . Smoothing is the one remaining idea it does not touch, because it adds no parameter at all. It only insists that the estimate at time t use the observations after t as well as those before.

The prediction is worth stating before the numbers, because it is the whole point. A smoother removes variance and lag. It cannot remove a bias in the measurement model: if every twilight reports the same distortion, propagating that information backwards as well as forwards reports the distortion twice. So a large gain is expected on the tuning window, where the Argos fixes are dense and the residual error is mostly spread, and little or none on the lockbox, where Section IX says the error is structural.

A. Two smoothers, and three things checked before trusting either

I implement both a forward-filtering backward-simulation smoother (FFBS) and the generalised two-filter smoother. They approximate the same marginals by different routes, so agreement between them is evidence and disagreement is a bug.

The forward pass is the configuration of Section VIII: hard seafloor set, SST aiding, 12 000 particles. To store its full state cloud I patch the source of the published filter rather than retyping its loop, so the two cannot drift apart; the resulting track is bit-identical to the published one (maximum absolute difference 0). The transition density used backwards is checked against 150 000 simulated draws of the forward propagation and agrees to 3.6×10^{-15} .

For the two-filter smoother the artificial prior on velocity is taken to be the stationary law of the Ornstein–Uhlenbeck velocity and the backward proposal its exact time reversal, which makes the importance ratio $\gamma_t(x_t)f(x_{t+1} | x_t)/[\gamma_{t+1}(x_{t+1})q(x_t | x_{t+1})]$ identically one. The code measures that ratio rather than trusting the algebra: the largest absolute log ratio over all 241 steps is 1.3×10^{-12} . Fusion is done between the particle clouds, not between Gaussian summaries of them.

TABLE IX

FILTER AGAINST SMOOTHER, FIVE SEEDS, MEDIAN GREAT-CIRCLE ERROR IN km, MEAN $\pm$ STANDARD ERROR. THE LAST COLUMN IS THE MEDIAN SIGNED NORTH ERROR ON THE LOCKBOX, WHICH IS WHAT SEPARATES VARIANCE FROM BIAS. THE TWO-FILTER ROW IS A SINGLE-SEED CROSS-CHECK ON THE SAME FORWARD PASS.

Estimate	Tune	Lockbox	Δ_N
Filter, mean	119.0 $\pm$ 1.4	404.2 $\pm$ 2.9	−378
FFBS, mean	68.6 $\pm$ 4.4	388.2 $\pm$ 22.8	−368
FFBS, geom. median	56.8 $\pm$ 3.8	412.7 $\pm$ 18.3	−380
Two-filter, mean	70.3	357.3	—

B. A large gain on the tuning window, none on the lockbox

Table IX gives the result over five seeds. On the tuning window the smoother cuts median error from 119.0 ± 1.4 to 68.6 ± 4.4 km, a gap of 50.3 km at 10.9 standard errors, and the geometric-median readout reaches 56.8 ± 3.8 km. This is the largest single improvement in the study, and it costs no new parameter and no new data.

On the lockbox the same smoother moves the error from 404.2 ± 2.9 to 388.2 ± 22.8 km, a gap of 0.7 standard errors, which is not a difference. The signed error is almost unchanged: 378 km too far south becomes 368 km too far south. The two smoothers also agree with each other to about 1 km on both windows despite sharing no backward machinery, which is the strongest single argument that the number is real.

Three controls support the reading. The answer is flat in the smoothing budget (66.5–67.3 km for K from 100 to 1600 trajectories), so it is not a compute artefact. None of the 96 400 smoothed states leaves the hard seafloor set, so admissibility survives smoothing by construction rather than by luck. And the posterior spread falls to 0.65 of the filter’s while 95% coverage on the tuning window *rises* from 92.2% to 93.8%: the intervals get narrower and more honest at the same time. On the lockbox coverage is 37.5% before and after, which is the signature of an interval centred in the wrong place rather than one that is too narrow.

One stated motivation does not survive. The reason usually given for smoothing here is lag, and there is very little lag to remove: shifting the filter’s track in time buys only 1.6 km, at a best shift of one day. The improvement comes from using future observations to sharpen each marginal.

C. A reason I had skipped this, and why it was wrong

An earlier step in the notebook rejected backward simulation on an explicit calculation: the position process noise over a median step is about 2 km while the posterior cloud is about 125 km across, so only ~ 3.5 of 12 000 particles lie within one standard deviation of a backward target. That arithmetic is correct and I reproduce it: the median effective number of ancestors per target is 2.9.

The conclusion drawn from it was not. Two claims were conflated. That the backward weights *underflow* is a statement about floating point, and it disappears once the weights are formed in log space with the row maximum subtracted. That

the sampler *collapses* does not follow at all, because each trajectory draws its own ancestor independently; a concentrated weight vector pins one trajectory rather than collapsing the ensemble. Measured directly, 306 of 400 smoothed trajectories remain distinct at the start of the record, and Table IX shows the ensemble is better calibrated than the filter it came from. Rejecting the method on that argument cost roughly 50 km of tuning-window accuracy.

XI. TEMPERATURE: TWO ACCIDENTAL CORRECTIONS

Section IX closes the twilight channel and Section X shows that a smoother removes variance but not bias. The one remaining place to look for independent information is the thermometer. The tag reports a surface temperature with each dawn, and the pipeline compares it to the NOAA OISST quarter-degree daily analysis through a Gaussian likelihood with one fitted bias and one fitted standard deviation, thinned to every third day to respect the analysis error correlation.

Two files in the same archive had never been opened: a binned depth–temperature profile roughly every two hours, and a mixed-layer summary. Neither supplies a second position fix. What they supply is a per-day statement of how far the one temperature reading can be trusted, and using them turned up a defect rather than an improvement.

A. The land mask was deleting the true position

Evaluating the residual at the *true* Argos position removes the position error and leaves the observation model exposed on its own. Done that way, the first thing visible is not a noise-model problem at all. On 19 of the 95 dawn events that carry both a temperature reading and a defensible truth—one fifth of them, and 14 of the 18 that fall in the lockbox—OISST returns no value at the animal’s own Argos fix. Seventeen of those nineteen are water according to the same bathymetry the seafloor constraint already uses, at a median depth of 103 m. They lie in the Gulf of California, which is narrower than the cell the analysis is built on.

The likelihood mapped a missing reference to $-\infty$, reasoning that the animal cannot be on land. A gap in OISST does not mean land; it means land to a quarter-degree product. Averaged over sampled positions on days when the term fires, it was assigning zero probability to 35.3% of the map, the true position included. Nothing is lost by treating a missing reference as uninformative instead: the hard seafloor set of Section VIII already rejects land exactly, at far finer resolution, and it is the term that is supposed to be doing that job.

B. The tag measures its own reliability

The published term uses one standard deviation for all 137 days. Measured against the truth on the tuning window the residual spread is 1.11°C where the filter assumes 0.69°C , so the temperature term is overconfident by a factor of 1.6. More useful than the average is that the tag predicts its own error. Sorted into quartiles of the tag’s own mixed-layer depth, the residual spread rises monotonically through 0.54, 0.75, 1.07 and 1.69°C ; a variance model linear in mixed-layer

TABLE X

FOUR VERSIONS OF THE TEMPERATURE TERM THROUGH THE SAME FORWARD FILTER, FIVE SEEDS, GREAT-CIRCLE ERROR IN km. STANDARD ERRORS ARE 0.5–3.8 km ON THE TUNING COLUMNS AND 1.7–10.6 km ON THE LOCKBOX MEAN. P75 IS THE 75TH PERCENTILE.

Temperature term	Med.	Mean	P75	Lock
As published	118.5	166.9	273.1	446.5
Missing cell uninformative	129.1	145.7	176.8	452.1
+ per-day σ	138.8	143.8	191.0	450.2
+ per-day bias	161.3	164.2	228.3	481.3

depth beats a constant one by $\Delta\text{AIC} = -19.9$ on the tuning window. Near-surface stratification does about as well and is uncorrelated with calendar time ($r = -0.04$), so this is water-column physics and not a drifting sensor. A competing hypothesis died cheaply: the animal reached the surface on every single day the term fires, so a story about the tag never getting shallow enough explains nothing.

The same geometry that makes a twilight bias unidentifiable applies here. The meridional OISST gradient along this track is 1.07°C per degree of latitude, so one degree of free thermometer bias buys 103 km of free latitude. A latent temperature-bias state is therefore the Section IX trap in a different channel. A bias regressed on an *observed* covariate adds no latent freedom and is the only safe form, and it is the only form tested below.

C. Repairing it rotates the error instead of shrinking it

Table X runs four versions of the temperature term through the same forward filter. Against the published term, treating missing cells as uninformative and taking the per-day standard deviation from the mixed-layer depth improves the tuning-window mean by 23 km, the RMS by 29 km and the 75th percentile by 82 km, each past ten standard errors—while making the median 20 km *worse*. Both statements are true, and which one is quoted is a decision about whether a typical day or a bad day matters more. This paper has quoted the median almost everywhere, and the median is precisely the statistic that cannot see a defect which ruins one day in five.

The signed error says why the median moved the wrong way. The median tuning-window offset goes from 33 km south and 70 km west under the published term to 97 km south and 13 km west under the fully repaired one, while its magnitude stays between 70 and 98 km. The error does not shrink; it rotates from south-west to very nearly due south. Two defects had been cancelling. Deleting every coastal cell acted as an offshore repulsion that nobody wrote down, worth about 25 km of eastward error. The unmodelled 0.58°C cold offset acted as a northward pull: at 1.07°C per degree of latitude it should be worth 60 km, and the observed loss of northward pull when it is modelled away is 64 km. Repairing the measurement model honestly removes both accidental corrections and exposes the southward twilight bias that Section IX says cannot be removed at all.

One further negative is worth recording. The per-day *bias* term is preferred by AIC on the observation residual and is worse than the per-day standard deviation alone on every filter

statistic in Table X. Fitting the residual better is not the same thing as locating the animal better, and an information criterion computed on the residual cannot distinguish them.

D. What the repair is worth once you smooth

Run through the smoother of Section X instead of the filter, the repair almost vanishes: tuning-window mean error moves from 94.4 ± 3.7 to 92.5 ± 1.2 km, half a standard error, with the 75th percentile improving by 14.7 ± 6.6 km and the lockbox mean going the wrong way by 20.2 ± 9.5 km. The filter needed the repair because it has only the past to work with, and a day on which the truth is deleted is unrecoverable. A smoother that can see both directions repairs those days by itself.

The conclusion for the temperature channel is therefore negative in the useful sense. It held no untapped accuracy. What it held was an unflagged defect that was making the filter wrong in a flattering direction, and the tag’s own depth–temperature profiles were enough to find it without a single new data source. The repair should be adopted anyway, on the grounds that a likelihood must not assign probability zero to a place the animal demonstrably was, and not on the grounds that it improves a score.

XII. DISCUSSION

A. What survives

A Student-*t* twilight likelihood is worth about 100 km and is the largest single modelling win in this work — considerably more than the choice of filter, which is worth about 26 km. Smoothing beats filtering consistently for all three estimators, and a particle smoother is the single largest improvement in the study: 119.0 ± 1.4 to 68.6 ± 4.4 km on the tuning window at 10.9 standard errors, with the posterior narrowing to 0.65 of its former spread while coverage improves (Section X). Independent aiding is what breaks the equinox degeneracy: SST cuts March error from 522 to 109 km, because its information is orthogonal to day length rather than because it is more accurate. And once a hard constraint makes the posterior multimodal, the readout stops being innocent: the same particle cloud reads 364 km at the posterior mean and 452 km at the local MAP. Against four alternatives on the same clouds — marginal mode, weighted geometric median, highest-weight particle and a single weighted draw — the posterior mean is nonetheless the best choice on the tuning window, and none of the four beats it out of sample by more than its seed noise.

B. What does not

The headline, as a measure of accuracy. The ablation ladder, which was built below the seed-noise floor. The raw-light recalibration, which improved the quantity it was fitted to and tripled the held-out error. The bias-state fix, as an accuracy claim, and in fact as any kind of estimate: Section IX shows the parameter was never identifiable from twilight data. And my own selection rule, as a sufficient safeguard — it is clean, and over a wider family it is actively anti-correlated with accuracy. And the temperature channel, as a reservoir of further accuracy: Section XI finds a real defect in it — the

satellite analysis has no value at the animal's own position on a fifth of the days the term fires, and the likelihood called those places impossible — but repairing that defect rotates the error rather than reducing it, and buys nothing that survives smoothing.

C. Limitations

This is one animal, one deployment, and one ground-truth track. The lockbox contains 20 matched events, so lockbox medians carry wide intervals — the seed standard deviation alone is 90 km, and sampling error on 20 events is comparable. Nothing here should be read as a general ranking of nonlinear filters; it is a statement about what this data set can resolve, which is less than I originally claimed. The Argos error model for classes 0, A and B is assumed rather than known. GPE3 is a black box, so I can say that it generalises better but not why.

D. What I would do differently

Hold out a block of ground truth before writing any model code — a block at the end, scored once. Report seed variance on every stochastic result and refuse to interpret any difference smaller than it; two seeds would have killed most of the ladder. And treat calibration support as a first-class check: for every fitted sensor parameter, compare the range of its regressors in the calibration window against the range expected at deployment. A goodness-of-fit plot cannot show you an unidentifiable parameter; a support check can. I would also be slower to reject a method on a back-of-envelope argument: I skipped backward simulation because a correct calculation about weight concentration was attached to an incorrect conclusion about ensemble collapse, and that one unchecked inference cost about 50 km (Section X). Above all, before adding any parameter to a state vector, count observations per epoch against unknowns per epoch: that takes a minute and would have told me that the whole of Section VIII was estimating a quantity the data cannot see. I would also report more than one error statistic. Almost every number here is a median, and a median cannot see a defect that ruins one day in five: the repair in Section XI improves the mean by 23 km and the 75th percentile by 82 km while making the median 20 km worse. And I would open every file in the archive before modelling rather than after; the evidence that the temperature term was broken had been sitting in two files the pipeline never read.

XIII. CONCLUSION

Light-level geolocation of this record is dominated not by the choice of nonlinear filter but by a biased measurement model that no filter in the family can detect from the data it is given. On the window I tuned on, my best configuration reaches 117 km and approaches the manufacturer's smoother. On 30 days I never looked at, it reaches 544 km against the smoother's 118 km, and reports a 201 km standard deviation while being wrong by 532 km.

The measurable contributions are therefore negative, and I think more useful than the positive claims they replace. Seed-to-seed spread of 29–90 km invalidates an ablation ladder built from 1–5 km steps. Tuning-window rank predicts held-out rank only weakly ($\rho = +0.30$), so the best-on-tuning variant ranks 54th of 75 out of sample. A recalibration that halved its own fitting residual tripled held-out error, because the parameter it adjusted was never identifiable from the calibration window. And a principled one-parameter fix, chosen without ever touching the held-out data, improved calibration and not accuracy — and could not have improved accuracy, because a twilight pair is two numbers, position is two unknowns, and a threshold bias is therefore a linear combination of the two things it was meant to correct. The same marginal-likelihood rule, given a three-dimensional bias family, prefers the degenerate direction in every one of its top configurations and lands a factor of four worse on the window it was computed on.

A smoother — the one repair identifiability does not forbid — then halves tuning-window error while leaving held-out error statistically untouched, which is the cleanest confirmation available that the residual failure is bias and not variance. The estimator that generalises here is the one I did not write. The part of this work I would defend is the protocol that let me find that out, and the part I would keep is the habit of checking whether a parameter is estimable before estimating it.

REPRODUCIBILITY

Everything runs top to bottom in a single Colab notebook with no manual steps. The tag archives are fetched at run time from the ATN Data Assembly Center by stable UUID, so a cold start gets byte-identical inputs; the raw data are not redistributed. All seeds are fixed in code. Particle counts are stated with every result: Table II uses 120,000 particles over ten seeds, and Tables III, V, VI and VIII use 20,000 particles over three seeds, and the readout comparison of Section IX uses five seeds. Table IX uses 12,000 forward particles over five seeds with 400 smoothed trajectories, and the two-filter cross-check adds 3,000 backward particles on one seed. The smoother carries two self-checks that run with it: the stored-cloud forward filter is asserted bit-identical to the published one, and the backward importance ratio is asserted to be one. Table X uses 12,000 particles over five seeds, and the smoothed comparison of Section XI adds 400 trajectories on the same five seeds; the modified temperature likelihood carries its own assertion, that it is numerically identical to the published one wherever the satellite analysis has a value, so that any change in the tracks is attributable to the missing cells alone. Every parameter of the temperature model is fitted on the tuning window only. Absolute values are therefore not comparable *between* those two groups, although every comparison *within* a table is like-for-like. Table I reports each configuration at the particle count at which it was built. Figures are regenerated by the notebook. Code, the technical report and build instructions are at <https://github.com/vivianesolomon/pat-tag-geolocation>.

Environment: Python 3.13, NumPy 2.1.3, SciPy 1.16.3, pandas 2.2.3, Matplotlib 3.10.0, netCDF4 1.7.4.

ACKNOWLEDGMENT

Tag data are published through the Animal Telemetry Network Data Assembly Center. The GPE3 reference tracks were produced by the tag manufacturer's software.